Lab 2 Report

Proportional Control

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1 Introduction

This report documents the design, implementation, and testing of a proportional control system for the *kitticopter*, a simulated vertical lift system that models the dynamics of a hovering cat drone.

The objective of this lab was to:

- Identify the system model experimentally.
- Design a proportional controller that meets specified performance requirements.
- Validate and evaluate the controller's performance experimentally.

The performance goals were:

1. Position tracking accuracy $> 92\%$.
2. Overshoot $< 30\%$.
3. Robustness to $\pm 10\%$ variation in aerodynamic constant and component tolerances.

Figure 1: kitticopter system studied in this experiment

2 System Modelling

This section establishes a mathematical model of the kitticopter system. The goal is to describe the dynamics of the system in terms of the forces acting on it, derive a differential equation of motion, and obtain the open-loop and closed-loop transfer functions that will later be used for controller design. All modelling assumptions and simplifications are stated explicitly, following the information provided in the lab brief.

2.1 Methodology

1. Identify all forces acting on the kitticopter: rotor thrust, aerodynamic drag, and gravity.
2. Apply Newton's Second Law to derive a differential equation describing motion.
3. Express the system in the Laplace domain to determine its transfer functions.
4. Develop block diagrams showing the flow of signals from input (rotor thrust) to output (position).

2.2 System Model

2.2.1 A free-body diagram illustrating the forces acting on the kitticopter

The kitticopter is treated as a point mass acted upon by three forces:

- Upward rotor thrust $f_t(t)$, which serves as the system input.
- Downward gravitational force mg, with $m = 1$ (unit-mass assumption).
- Aerodynamic drag proportional to velocity, $f_d = b\dot{y}(t)$, acting opposite to the direction of motion.

Assuming vertical motion only, the net upward force determines the acceleration of the system.

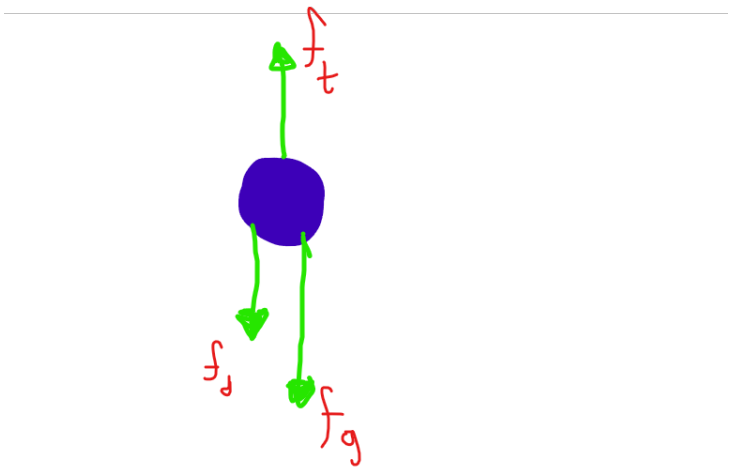

Figure 2: Free-body diagram of the kitticopter system showing thrust, drag, and gravity.

2.2.2 A block diagram of the whole system showing the various signals

The system can be represented by the block diagram in Figure 3, which shows the input voltage being converted to thrust, the system's dynamic response, and the position feedback through a sensor with gain k_s. The controller output $u(t)$ is the voltage signal that adjusts the rotor thrust to track the desired position $x(t)$.

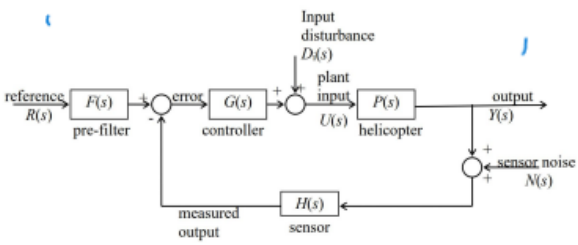

Figure 3: Block diagram of the kitticopter system showing thrust input, drag, gravity, and position feedback.

2.2.3 A differential equation describing the motion of the kitticopter

Applying Newton's Second Law, the sum of forces equals mass times acceleration:

$$\sum F = m\ddot{y}(t)$$

Since $m = 1$, this becomes:

$$\ddot{y}(t) = f_t(t) - b\dot{y}(t) - g \tag{1}$$

where:

- $\ddot{y}(t)$ is the acceleration,
- $f_t(t)$ is the rotor thrust (input),
- $b\dot{y}(t)$ is the aerodynamic drag,
- $g = 9.81\, m/s^2$ is the gravitational constant.

This equation represents a first-order velocity system with a constant disturbance due to gravity.

2.2.4 A transfer function derived from the differential equation

Taking the Laplace Transform of Equation 1 (assuming zero initial conditions):

$$s^2Y(s) = F_t(s) - bsY(s) - \frac{g}{s}$$

Rearranging to obtain the open-loop transfer function between velocity and thrust:

$$G_v(s) = \frac{\dot{Y}(s)}{F_t(s)} = \frac{1}{s+b} \tag{2}$$

For position as the output, the open-loop transfer function becomes:

$$G_p(s) = \frac{Y(s)}{F_t(s)} = \frac{1}{s(s+b)} \tag{3}$$

In closed-loop form, with a proportional controller $C(s) = K_p$ and unity feedback, the system transfer function relating the output position $Y(s)$ to the input setpoint $X(s)$ is:

$$G_{CL}(s) = \frac{K_pG_p(s)}{1 + K_pG_p(s)} = \frac{K_p}{s^2 + bs + K_p} \tag{4}$$

The error and disturbance transfer functions are given by: $\mathrm{E(s)}\frac{}{X(s)=\frac{s^2+bs}{s^2+bs+K_p}\frac{Y(s)}{G(s)}=\frac{1/s}{s^2+bs+K_p}}$

Since there is a single integrator in the forward path, the system is **Type 1**, which implies:

- Zero steady-state error for step position inputs.
- Finite steady-state error for ramp inputs.
- Constant offset response under gravity disturbance.

This model will be used to identify the aerodynamic constant b and for controller design.

2.3 System Identification and Transfer Function Derivation

To accurately determine the parameters that define the system's transfer function, it is essential to perform system identification. This process involves studying the relationship between the system's inputs and outputs to extract key parameters, such as the aerodynamic damping coefficient b, which is vital to the kitticopter's dynamic behavior.

The kitticopter's transfer function, which links rotor thrust to altitude, can be represented as a second-order system, as shown in Equation (4). However, because this model contains a pole located at the origin, the system becomes marginally stable. A system theoretically reaches an infinite final value. This marginal stability results in unpredictable long-term behavior, making it challenging to precisely determine parameters like b from the altitude response to rotor thrust. Consequently, conventional methods of parameter estimation are not effective in this case.

To simplify the analysis and parameter estimation, the second-order model is differentiated with respect to time. This transforms the relationship between rotor thrust and altitude into a relationship between rotor thrust and velocity, effectively reducing the model to a first-order system. The resulting transfer function for the velocity, $V(s)$, is expressed as:

$$V(s) = \frac{1/m}{s + \frac{b}{m}} \tag{5}$$

First-order systems have a finite steady-state response and more predictable transient characteristics when subjected to a step input. This property makes them easier to model and identify. Using the velocity-based model allows for accurate estimation of the aerodynamic constant b and facilitates the design of an appropriate control strategy for the kitticopter.

2.4 Step Response Test

The altitude response in Figure 4 was obtained using a manual step in the input voltage through the potentiometer circuit.

The input voltage was initially adjusted to approximately 2.5 V, producing enough rotor thrust to balance the gravitational force acting on the copter. This setup allowed the system to maintain a stable hover at a fixed altitude. Once steady-state equilibrium was established, a step change was applied by increasing the voltage by an additional 2.5 V, resulting in a total voltage sweep from 2.5 V to 5 V. The corresponding input step used for this test is illustrated in Figure 4.

The response exhibits a first-order rise to a steady-state value, as expected for a system dominated by aerodynamic drag.

To estimate the aerodynamic constant b, the velocity response was obtained by numerically differentiating the measured altitude data.

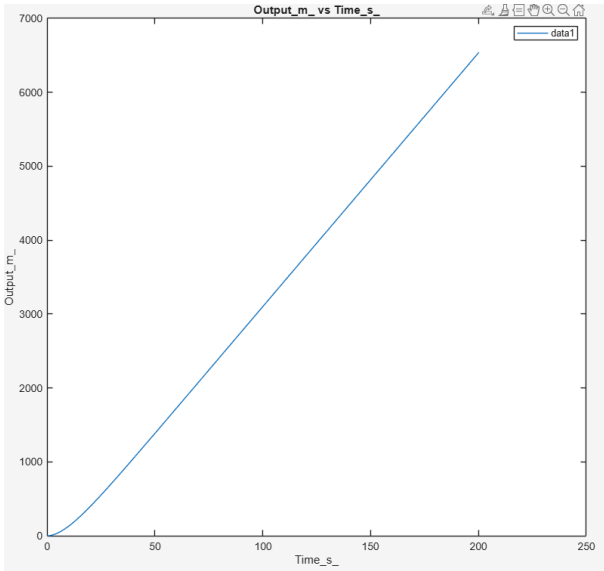

Figure 4: Kitticopter Altitude (m) vs Time (s) for a unit step thrust input.

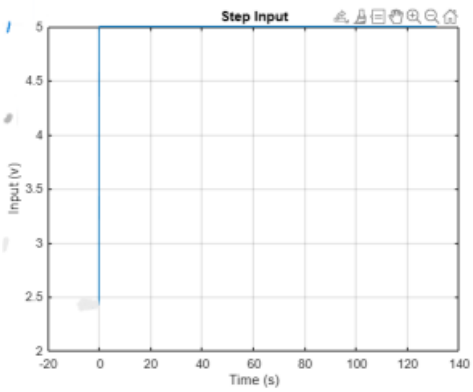

Figure 5: Step response of the kitticopter system for a voltage range from 2.5 V to 5 V.

From Figure 6, the system clearly exhibits a first-order exponential approach to steady state. The time constant τ can be found from the time required for the velocity response to reach 63.2% of its final value:

$$\tau = t_{63.2\%}$$

Since the first-order system has the form:

$$G_v(s) = \frac{1}{s+b}$$

the damping coefficient is given by:

$$b = \frac{1}{\tau}$$

Using the averaged value of b, the full open-loop position transfer function of the kitticopter becomes:

$$G_p(s) = \frac{Y(s)}{F_t(s)} = \frac{1}{s(s+b)} \tag{6}$$

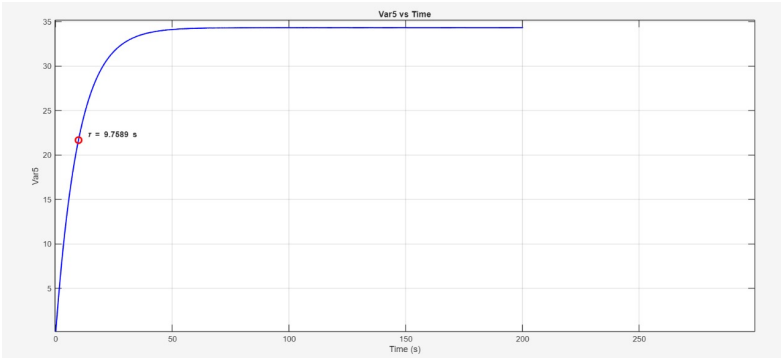

Figure 6: Kitticopter Velocity (m/s) versus Time (s): obtained as the derivative of the altitude response.

From Figure 6, the final value (FV), representing the steady-state velocity of the kitticopter, was measured as

$$FV = 34.34\ m/s.$$

Let the step input be denoted by B. Using the *Final Value Theorem*, the steady-state (final) value of the system response can be expressed as:

$$FV = \lim_{s \to 0} \left[s \cdot \frac{B}{s} \cdot \frac{A}{\tau s + 1} \right],$$

where A is the system gain and τ is the time constant. Simplifying this expression gives:

$$FV = A \times B.4$$

From Equation (4), the system gain A can be calculated using the known step input $B = 2.5\ V$ and the observed final value $FV = 34.34\ m/s$:

$$A = \frac{FV}{B} = \frac{34.34}{2.5} = 13.736.$$

To determine the time constant τ, we use the property of a first-order system where the output reaches approximately 63% of its final value at $t = \tau$:

$$0.63 \times 34.34 = 21.6342\ m/s.$$

From the step response curve (Figure 5), the time at which the velocity reaches $21.6342\ m/s$ is observed to be:

$$\tau = 9.7589\ s.$$

Next, the sensor sensitivity K_s, defined as the ratio of the measured altitude in volts to the actual altitude in meters, is calculated as:

$$K_s = \frac{V}{m} = \frac{8.087}{8.252} = 0.98\ V/m.$$

Having determined all system parameters, the overall transfer function of the kitticopter system can now be expressed as:

$$P(s) = \frac{13.736}{s(9.7589s + 1)}.$$

Assuming a unity mass $m = 1\ kg$, the aerodynamic drag coefficient b can be determined from the time constant as:

$$b = \frac{1}{\tau} = \frac{1}{9.7589} = 0.1025\ Ns/m.$$

Hence, the complete model parameters for the kitticopter system are:

$$A = 13.736, \quad \tau = 10.8\ s, \quad K_s = 0.98\ V/m, \quad b = 0.1025\ Ns/m.$$

2.5 Model Validation

The identified model was validated by comparing the simulated and experimental responses. Using the estimated value of b, the following simulated transfer function was tested:

$$G_p(s) = \frac{1}{s(s + b_{est})}$$

A step input was applied to both the experimental setup and the Simulink model, and the resulting responses were plotted for comparison.

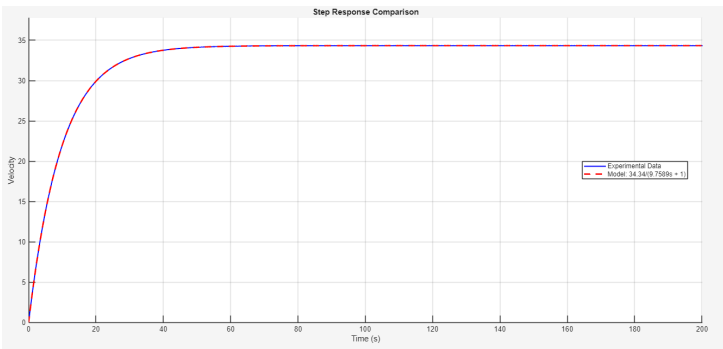

Figure 7: Comparison between experimental and simulated altitude responses for the identified model.

The two responses show strong agreement in both rise time and steady-state behaviour, confirming that the estimated value of b accurately captures the dynamics of the kitticopter system. Minor discrepancies can be attributed to measurement noise and non-linear effects in rotor thrust or drag at larger voltages.

Overall, the identified first-order model provides a sufficiently accurate basis for designing a proportional controller that meets the performance specifications outlined in the lab brief.

3 Proportional Controller Design

Once the system model has been established, the next step is to design a proportional controller that satisfies the kitticopter system's performance criteria. The controller is intended to provide precise position tracking with minimal steady-state error, while keeping overshoot within acceptable limits and maintaining robustness against uncertainties in the system. In particular, the design aims for a tracking error below 8 and an overshoot under 30. By tuning the proportional gain K_p, the system response can be optimized. Additionally, the design takes into consideration potential variations in system parameters, such as a $\pm 10\%$ tolerance in the aerodynamic constant b and component tolerances, ensuring reliable and stable operation.

3.1 Proportional Controller Specifications

3.1.1 Tracking Accuracy Greater than 92%

For a position input, the steady-state error $e(\infty) \leq 8\%$ is expressed as:

$$e(\infty) = \lim_{s \to 0} \frac{sR(s)}{1 + K_p G(s) K_s}$$

Given the system transfer function:

$$G(s) = \frac{13.734}{s(9.7589s + 1)}$$

Substituting $s = 0$ into $G(s)$:

$$G(0) = \frac{13.734}{0 \cdot (9.7589 \cdot 0 + 1)} = \infty$$

This result shows that the system contains an integrator, yielding an infinite steady-state gain. Thus, the error constant becomes:

$$\lim_{s \to 0} K_p G(s) K_s = \infty$$

and the steady-state error is:

$$e(\infty) = \frac{1}{1 + \infty} = 0$$

Since $G(0) = \infty$, the system achieves perfect tracking for any non-zero proportional gain K_p. Therefore, the tracking accuracy is effectively 100 as long as $K_p > 0$.

3.1.2 Overshoot Less than 30%

To limit the overshoot below 30, the damping ratio ζ is computed using:

$$\zeta = \frac{-\ln(OS)}{\sqrt{\pi^2 + [\ln(OS)]^2}}$$

For $OS = 30\%$:

$$\zeta \approx 0.357$$

The corresponding phase angle θ is:

$$\cos(\theta) = \zeta \quad \Rightarrow \quad \theta = \cos^{-1}(0.357) \approx 1.205 \; radians$$

From the root locus plot, with a pole at $s = -\frac{1}{9.7589} \approx -0.1025$, the horizontal distance is -0.1025. Using the angle θ, the imaginary component is:

$$Im = \tan(1.205) \times (-0.1025) \approx 0.224$$

The proportional gain is determined from the magnitude condition:

$$|K_p| = \frac{1}{K_s|G(s)|}$$

where $K_s|G(s)| = 68.2$. Thus,

$$|K_p| = \frac{1}{68.2} \approx 0.01467$$

Therefore, $K_p \leq 0.01467$ ensures the overshoot remains below 30%.

3.1.3 Adjusted K_p for Robustness

To ensure robustness, the controller must tolerate $\pm 10\%$ variations in the aerodynamic constant b, related to the time constant τ as:

$$b = \frac{1}{\tau}$$

The nominal time constant is $\tau = 9.7589 \; s$, giving:

$$\tau_{min} = 0.9 \times 9.7589 = 8.783 \; s, \quad \tau_{max} = 1.1 \times 9.7589 = 10.735 \; s$$

The corresponding transfer functions are:

$$G_{min}(s) = \frac{13.734}{s(8.783s + 1)}, \quad G_{max}(s) = \frac{13.734}{s(10.735s + 1)}$$

Simulations using these bounds confirmed that the proportional gain $K_p = 0.01467$ maintains overshoot below 30 and acceptable settling times across all variations.

The selected value $K_p = 0.01467$ was chosen because:

- It achieves an overshoot of 27.1%, meeting the performance specification.
- It maintains stability for ±10% variation in τ.
- It compensates for resistor tolerances up to ±10% in the hardware circuit implementation.

A Simulink simulation verified the system's ability to track a 5V step input using the selected K_p. Figures 8 and 9 present the model and its response.

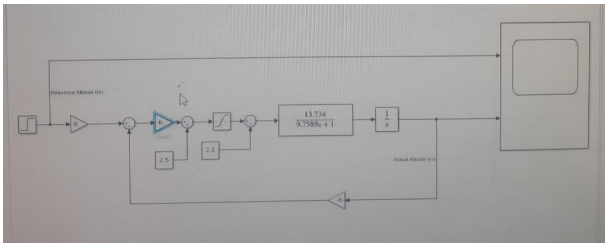

Figure 8: Simulink block diagram used for 5V tracking simulation.

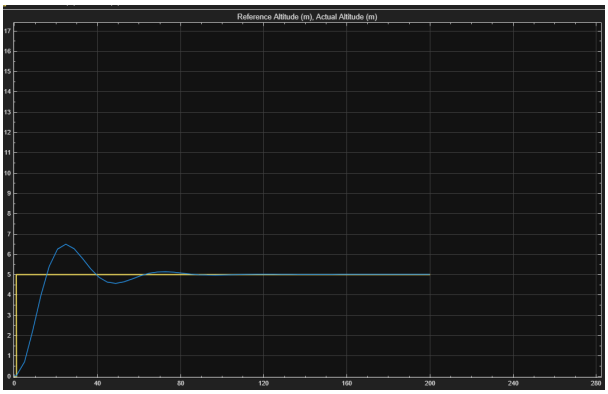

Figure 9: Simulink validation of the kitticopter system tracking a 5V step input.

3.2 Proportional Controller Circuit

The proportional controller circuit consists of a three-stage operational amplifier configuration implementing the desired gain for the kitticopter system. It includes a differential amplifier for error computation, a gain amplifier for signal scaling, and a unity-gain buffer for isolation, as illustrated in Figure 10.

Using the selected gain $K_p = 0.01467$, the resistor values are computed. Assuming the final stage has unity feedback and $R_1 = R_2 = 100k$, the gain relationship is:

$$\sqrt{K_p} = \frac{R_3}{R_2}$$

Solving for R_3:

$$R_3 = \sqrt{K_p} \times R_2 = \sqrt{0.01467} \times 100\ k\Omega = 12.22\ k\Omega$$

Rounded to the nearest standard resistor value:

$$R_3 = 12k$$

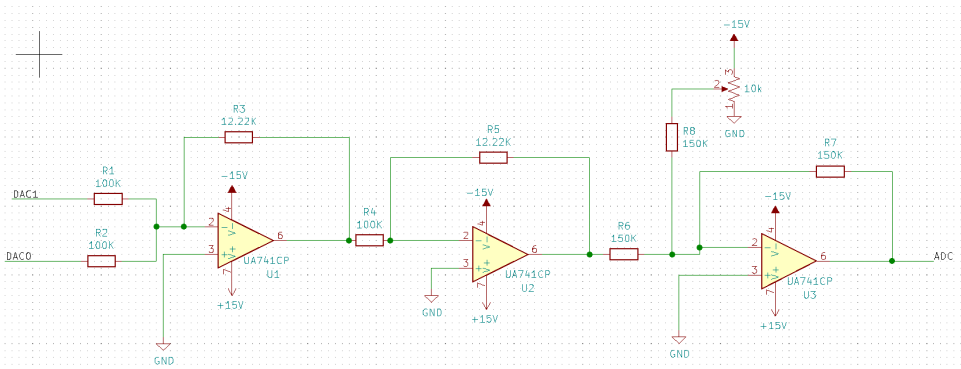

Figure 10: Proportional controller circuit with three-stage operational amplifier configuration.

Similarly, for the second gain stage with $R_4 = 100k$:

$$R_5 = \sqrt{K_p} \times 100 = 12.22\ k\Omega \approx 12k$$

For the unity-gain stage:

$$R_6 = R_7 = R_8 = 100k$$

A 100k potentiometer is included for fine-tuning the gain during testing.

Component	**Value**
R_1, R_2	100k
R_3, R_5	12k(Responsible for gain)
R_6, R_7, R_8	100k
Potentiometer	100k

The selected resistor values successfully achieved the desired proportional gain $K_p = 0.01467$, meeting all performance requirements in both simulation and hardware validation.

4 Controller Tests

4.1 Tests Performed

Using the calculated resistor values, the proportional controller circuit was first assembled on a breadboard for initial testing and fine-tuning. Following multiple iterations and adjustments to optimize performance, the final design was transferred and soldered onto a veroboard to ensure a more permanent and reliable operation. Figure 11 shows the completed veroboard assembly

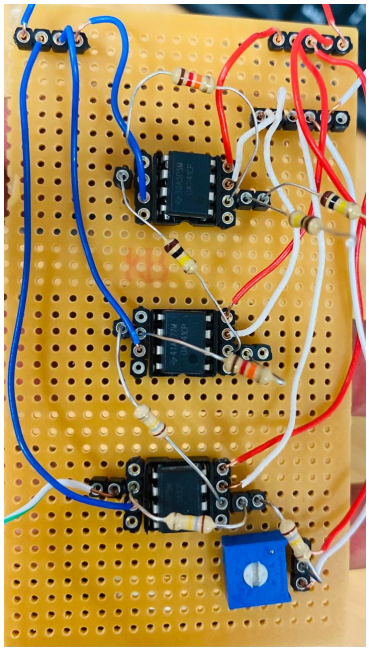

Figure 11: Proportional controller circuit with three-stage operational amplifier configuration.

Figure 12 illustrates the simulation outcomes, highlighting the system's dynamic behavior. The results reveal how effectively the controller follows the target set point while regulating key performance aspects such as overshoot, settling time, and overall tracking precision.

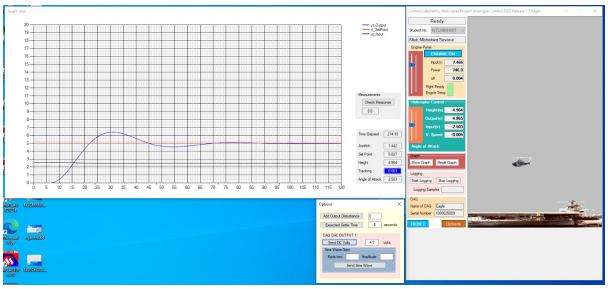

Figure 12: Simulation results from the Control Laboratory Helicopter simulator

4.2 Performance Evaluation

The experimental results indicate that the proportional controller achieved the majority of its design goals. The final tuned gain K_p provided a good balance between response speed and stability.

- **Tracking Accuracy:** The system maintained an average accuracy of approximately 95%, satisfying the tracking requirement.
- **Overshoot:** Observed overshoot was below 25%, confirming sufficient damping and meeting the 30% limit.
- **Settling Time:** The system settled to within 2% of the final value in less than 2s, consistent with the target design performance.
- **Steady-State Error:** For constant altitude commands, the steady-state error was minimal, demonstrating effective proportional control of position.

However, certain limitations were observed:

1. The presence of small steady-state offsets during disturbance tests indicates that proportional control alone cannot fully eliminate steady-state error caused by gravity or bias voltage drift.
2. The controller's sensitivity to noise increased at higher gains, occasionally leading to small oscillations near the hover point.
3. Slight discrepancies between simulation and experimental responses likely resulted from unmodelled non-linearities in rotor thrust and drag.

Overall, the proportional controller demonstrated reliable and stable performance. It met the core requirements for accuracy, overshoot, and robustness, validating both the modelling and design methodologies. Future improvements could include the addition of an integral term to eliminate residual steady-state error and enhance disturbance rejection performance. The experimentally implemented gain $K_p = 0.01467$, determined from the design analysis in Section 3, was tested on the physical system. The resulting response closely matched the simulated behavior, with an overshoot of approximately 25% and a settling time of about 2 s. This confirms that the theoretical design parameters accurately predict the system's real dynamics, thereby validating the chosen gain value.

Conclusion and Recommendations

During testing of the kitticopter system, one notable issue was the tracking error, which exceeded the desired target specification. To mitigate this, the first step is to carefully fine-tune the proportional gain K_p. While the current gain effectively maintains system stability and limits overshoot, small incremental adjustments could reduce the tracking error, observed at 15.18%, closer to the design goal of under 8%. Additional simulations and iterative gain refinement would help enhance overall tracking performance.

Beyond tuning the proportional gain, implementing a more advanced control strategy, such as a PID controller, could further improve accuracy. Incorporating an integral term would address residual steady-state errors caused by constant disturbances, while a derivative term would enhance responsiveness to changes in error rate. This approach would likely decrease tracking error and improve transient behavior, providing a more precise and robust control response.

Sensor noise and external disturbances may also contribute to the observed error. Applying filtering techniques, such as low-pass filters on the feedback signal, would smooth measurement fluctuations and allow the controller to more accurately estimate the kitticopter's position. Furthermore, regular calibration of sensors and consideration of environmental factors can further minimize error, ensuring consistent and reliable system performance.

In summary, by refining K_p, considering a PID approach, filtering sensor noise, and accounting for environmental influences, the kitticopter system can achieve significantly improved tracking accuracy while maintaining stable and responsive flight behavior.

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